F70TS: TIME SERIES AND MACHINE LEARNING

Time Series Coursework 1 2025/26

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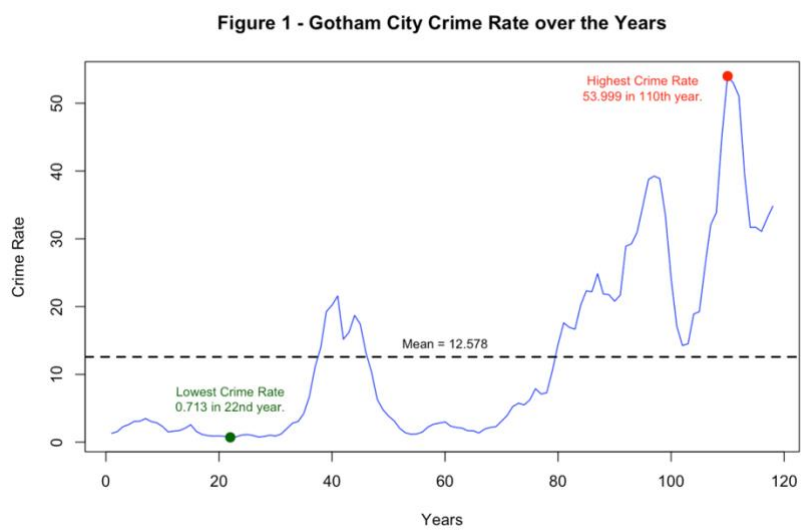
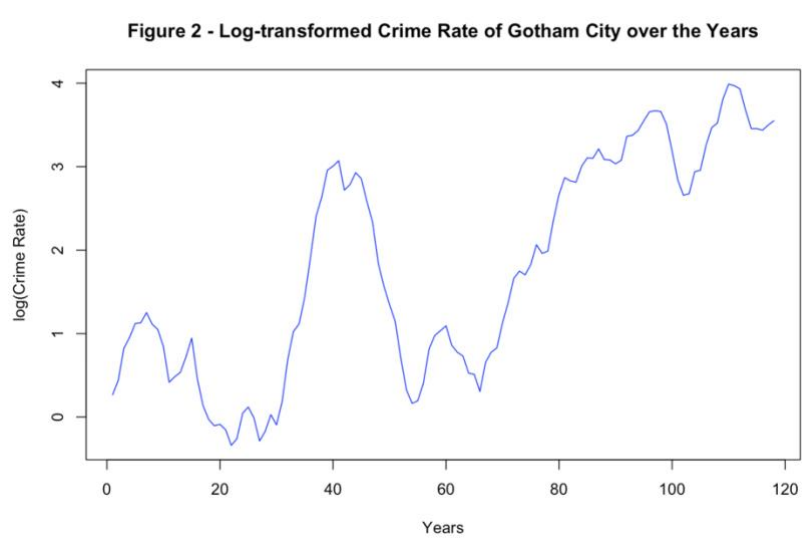
Time Series Analysis of Crime Rates in Gotham

1. The crime rate data of Gotham is shown in **Figure 1**. The crime rate has seen an overall upward trend in the time period of 118 years. This suggests that the mean is not constant. The crime rate varies with non-constant variance, indicating non-stationarity of the crime rate data.

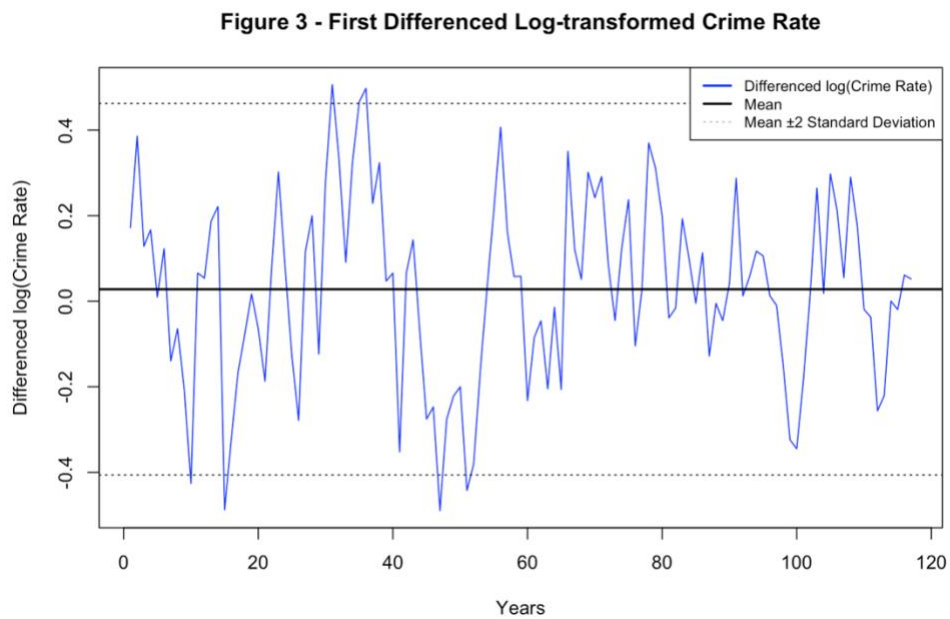
   Through R computation, we observe that the 22nd year saw the lowest crime rate at 0.713, and the 110th year saw the highest crime rate at 53.999. The mean crime rate over the 118 years is 12.578. The median crime rate is 5.387. The standard deviation of the data is 13.615.



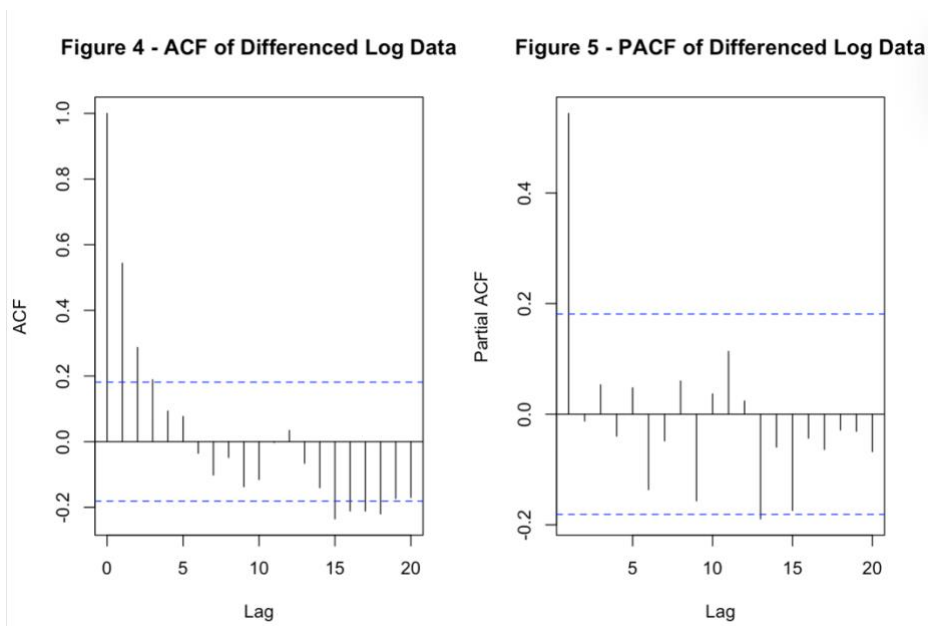
2. The log-transformed Crime Rate Data is taken to stabilize variance. It is plotted in **Figure 2**. Although the variance has been slightly stabilized, the upwards trend persists. The mean and variance are still non-constant. Therefore, the log-transformed data is non-stationary.



3. **Figure 3** shows the first-order differenced and log-transformed data. There does not seem to be an overall trend with this data. The data also does not seem to vary unstably. It seems to have attained **weak stationarity** since the data has a fairly constant mean and variance. The mean (black solid line) and mean $\pm 2 \times$ S.D. (grey dotted lines), which is considered a safe range for approximately constant variance, are plotted in the figure.



4. **Figure 4** and **Figure 5** show the sample ACF and sample PACF of the differenced and transformed crime data up to 20 lags, respectively. The figures show that both the ACF and the PACF of the differenced log-transformed data decay within a few lags. The decay is not gradual, which points to the stationarity of the differenced log-transformed data. The ACF cuts off after lag 3 and the PACF cuts off after lag 1. The ACF decays more slowly compared to the PACF of the differenced log-transformed data. This indicates that the differenced log-transformed data might follow an AR process.



5. To propose an ARIMA(p,d,q) model for the log-transformed crime data, we observe **Figure 4** and **Figure 5.** In particular, we consider the PACF plot when identifying an AR model and the ACF plot when identifying an MA model to fit to the data (F70TS Lecture Notes).

   We consider different models as per our observations and compare them to propose the final ARIMA(p,d,q) model.

   From Question 3, we know that the log-transformed crime data achieved stationarity after being differenced once. Therefore, d = 1.

   Figure 4 shows that the ACF cuts off after lag 3. Hence, we consider Model 1: ARIMA(0,1,3).

   Figure 5 shows that the PACF cuts off after lag 1. Hence, we consider Model 2: ARIMA(1,1,0).

   Model 2: ARIMA(1,1,0) is proposed as the better model to fit the log-transformed crime data for the following reasons:

   a. The ACF decays rather gradually compared to the PACF which exhibits a sharp cut-off after lag 1. This is characteristic of an AR(1) process.
   b. The AIC of Model 2 (-61.245) is less than the AIC of Model 1 (-57.28), which indicates that Model 2 is a better fit for the data. See R output in **Figure 6.**
   c. The BIC of Model 2 (-52.959) is less than the BIC of Model 1 (-43.469), which indicates that Model 2 is a better fit for the data. See R output in **Figure 6.**
   d. Model 2 has less number of parameters than Model 1, which is favorable due to its parsimony.

Figure 6 – AIC and BIC of Model 1 and Model 2

```
> AIC(model1)
[1] -57.28012
> AIC(model2)
[1] -61.24523
```

```
> BIC(model1)
[1] -43.46925
> BIC(model2)
[1] -52.95871
```

6. Fitting the proposed Model 2: ARIMA(1,1,0) assuming non-zero mean, we find the AIC to be -61.245 (**Figure 6**). This AIC value is very low, indicating that the model is a good fit for the log-transformed crime data.

7. We now write down our fitted Model 2: ARIMA(1,1,0) in the form:

$$\left(1-\phi_1 B-\cdots-\phi_p B^p\right)\left((1-B)^d \log(X_t)-\mu\right)=\left(1+\theta_1 B+\cdots+\theta_q B^q\right)\varepsilon_t,\ \varepsilon_t \sim WN(0,\sigma^2)$$

   Assumptions:
   X_t is the Gotham crime rate data.
   B is the backshift operator.

We employ the argument include.drift of the Arima( ) command to assume a non-zero mean, when d=1. Thus, μ (estimated mean of $(1-B)^1 \, log \, X_t$), becomes equal to the drift.

See R output in **Figure 7** for the parameters below:

$\phi_1 = 0.541 \quad \sigma^2 = 0.033 \;\; \mu = \; 0.030$

Figure 7 – Model 2: ARIMA(1,1,0) Summary

```
> summary(model2)
Series: log_crime
ARIMA(1,1,0) with drift

Coefficients:
         ar1   drift
      0.5414  0.0297
s.e.  0.0770  0.0362

sigma^2 = 0.03343:  log likelihood = 33.62
AIC=-61.25   AICc=-61.03   BIC=-52.96

Training set error measures:
                        ME      RMSE       MAE      MPE     MAPE      MASE        ACF1
Training set -0.0008421001 0.1804956 0.1417939 27.44872 46.29654 0.8158105 0.007519906
```

Let Y_t be the differenced log-transformed data

$$Y_t \coloneqq (1-B)\log(X_t)$$

Since Y_t follows ARIMA(1,1,0) process,

$$Y_t = \phi_1 Y_{t-1} + \varepsilon_t, \, \varepsilon_t \sim WN(0, \sigma^2)$$

Since Y_t has a non-zero mean, $Y_t - \mu \sim$ ARIMA(1,1,0).

$$(Y_t - \mu) = \phi_1 (Y_{t-1} - \mu) + \varepsilon_t$$

$$\Longrightarrow (Y_t - \mu) = \phi_1 B(\mathrm{Y_t} - \mu) + \epsilon_t$$

$$\Longrightarrow (1 - \phi_1 B)(Y_t - \mu) = \varepsilon_t$$

Substituting back $Y_t \coloneqq (1-B)\log(X_t)$,

$$(1 - \phi_1 B)\big((1-B)\log(X_t) - \mu\big) = \varepsilon_t, \quad \varepsilon_t \sim WN(0, \sigma^2)$$

Substituting the estimated parameters:

$$(1 - 0.541B)\big((1-B)\log(X_t) - 0.030\big) = \varepsilon_t, \; \varepsilon_t \sim WN(0{,}0.033)$$

which is of the required form,

$$\left(1-\phi_1 B-\cdots-\phi_p B^p\right)\left((1-B)^d \log(X_t)-\mu\right)=\left(1+\theta_1 B+\cdots+\theta_q B^q\right)\varepsilon_t,\ \varepsilon_t \sim WN(0,\sigma^2).$$

8. The model proposed by Batman is Model 3: ARIMA(2, 1, 1). The model is fitted to the log-transformed crime rate data. See **Figure 8** for the summary.

Figure 8 – Model 3: ARIMA(2,1,1) Summary

```
> summary(model3)
Series: log_crime
ARIMA(2,1,1) with drift

Coefficients:
          ar1     ar2     ma1   drift
      -0.1125  0.3240  0.6786  0.0293
s.e.   0.3698  0.2253  0.3503  0.0353

sigma^2 = 0.03391:  log likelihood = 33.81
AIC=-57.61   AICc=-57.07   BIC=-43.8

Training set error measures:
                        ME      RMSE       MAE      MPE     MAPE      MASE        ACF1
Training set -0.0006913247 0.1802041 0.1414286 26.63483 45.16295 0.8137091 -0.01258297
```

Model 2: ARIMA(1,1,0) is proposed as the better model to fit the log-transformed crime data for the following reasons:

a. The AIC of Model 2 (-61.245) is less than the AIC of Model 3 (-57.615), which indicates that Model 2 is a better fit for the data (**Figure 9**).
b. The BIC of Model 2 (-52.959) is less than the BIC of Model 3 (-43.804), which indicates that Model 2 is a better fit for the data (**Figure 9**).
c. Model 2 has less number of parameters than Model 3, which is favorable due to its parsimony.

Figure 9 – AIC and BIC of Model 3 and Model 2

```
> AIC(model3)    > BIC(model3)
[1] -57.61452    [1] -43.80365
> AIC(model2)    > BIC(model2)
[1] -61.24523    [1] -52.95871
```

References

F70TS Lecture Notes